

## **Vacuum Cleaner Problem**

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### **Aim**

To implement the **Vacuum Cleaner Problem** using a simple reflex agent in Python.

### **Algorithm**

1. Start.
2. Define the environment with two rooms (A and B).
3. Place the vacuum cleaner in one of the rooms.
4. Check the current room.
5. If the room is dirty, clean it.
6. Otherwise, move to the other room.
7. Repeat until both rooms are clean.
8. Display the final state.
9. Stop.

### **Code**

```
# Vacuum Cleaner Problem
```

```
# Initial state of rooms
```

```
rooms = {  
    'A': 'Dirty',  
    'B': 'Dirty'  
}
```

```
# Vacuum starts in Room A
```

```
location = 'A'
```

while True:

```
    print("Vacuum is in Room", location)
```

```
    if rooms[location] == 'Dirty':
```

```
        print("Cleaning Room", location)
```

```
        rooms[location] = 'Clean'
```

```
    else:
```

```
        print("Room", location, "is already Clean")
```

```
    # Check if all rooms are clean
```

```
    if rooms['A'] == 'Clean' and rooms['B'] == 'Clean':
```

```
        print("\nAll rooms are clean.")
```

```
        break
```

```
    # Move to the other room
```

```
    if location == 'A':
```

```
        location = 'B'
```

```
    else:
```

```
        location = 'A'
```

```
print("\nFinal State:")
```

```
print(rooms)
```

## OUTPUT:

```
>>> ===== RESTART: C:/Users/sri_jayu/Downloads/vaccum cleaner.py =====
Vacuum is in Room A
Cleaning Room A
Vacuum is in Room B
Cleaning Room B
All rooms are clean.
Final State:
{'A': 'Clean', 'B': 'Clean'}
>>>
```

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**RESULT:** Thus, the Vacuum Cleaner Problem was implemented successfully using a Simple Reflex Agent, and both rooms were cleaned.